



HITESH RAM VANDANAPU

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PROFILE

Computer Vision & Sensor Fusion Engineer with hands-on experience developing real-time perception pipelines using RGB cameras, LiDAR, and 3D data. Skilled in C++, Python, ROS 2, 3D geometry, point-cloud processing, sensor fusion, data association, Kalman filter, and state estimation. Experienced in building and validating perception systems on real-world sensor data for perception system and robotics applications, with a strong foundation in spatial estimation, multi-object tracking, and computer vision.

EXPERIENCE

Master's Thesis – 2D/3D Multi-Sensor Perception

Feb 2026 – Jul 2026

Technische Hochschule Ingolstadt (THI), Fraunhofer IVI

Ingolstadt, Germany

- Developed a multi-sensor perception and state-estimation framework combining 3D spatial data with RGB camera streams for real-time 3D trajectory tracking at 10 fps.
- Implemented LiDAR-camera fusion, 3D-to-2D projection, data association, and Kalman-filter-based state estimation.
- Investigated motion models including CTRV, CTRA, CV, and CA to evaluate observability, motion degeneracy, and estimation performance across different vulnerable road-user profiles.
- Designed and evaluated the complete perception pipeline on 22 real-world scenes, balancing computational latency and tracking accuracy to achieve a best pedestrian HOTA of 67.04%.

Research Intern – Computer Vision & Perception

Aug 2025 – Jan 2026

Fraunhofer Institute for Transport and Infrastructure Systems IVI

Ingolstadt, Germany

- Developed C++ and Python workflows for synchronizing, organizing, and validating LiDAR and camera streams from real-world ROS bag recordings for multi-modal perception pipelines.
- Transformed and merged point clouds from two LiDAR sensors using extrinsic transformations, then applied camera intrinsic and extrinsic parameters for 3D-to-2D projection across three RGB cameras.
- Processed and curated real-world datasets for point-cloud processing, visualization, sensor-data validation, and evaluation of autonomous perception algorithms.
- Retrained and deployed a YOLO11s object-detection model in a real-time roadside perception system, achieving 96.7% mAP@0.5.
- Optimized perception workflows for high-bandwidth streaming data, connecting research datasets with real-time execution and field-data validation.

Programmer Analyst Trainee

Apr 2022 – Oct 2022

Cognizant Technology Solutions

Chennai, India

- Supported software development, debugging, testing, and system maintenance while collaborating with cross-functional engineering teams to resolve technical issues.

Intern

Cognizant Technology Solutions

May 2021 – Oct 2021

Chennai, India

- Completed software engineering training focused on backend development, debugging, Java, SQL, and relational database management.

PROJECTS

Material Estimation from Images

- Developed an image-based material estimation pipeline using image segmentation and optimization under controlled lighting conditions.
- Used thresholding for initial parameter estimation and the Adam optimizer to refine parameters of the Disney BRDF model.
- Evaluated GPU and CPU execution approaches using CUDA and LLVM, demonstrating the impact of computational architecture on real-time processing.

Crack Detector – U-Net Image Segmentation

- Developed a TensorFlow-based U-Net segmentation pipeline for detecting structural cracks in wall-surface imagery.
- Applied image normalization and data augmentation and optimized the model using Adam, achieving 92% accuracy.

EDUCATION

Master of Science in Digital Engineering

Bauhaus University Weimar

Nov 2022 – Jul 2026

Weimar, Germany

Relevant Coursework: 3D Geometry, Photogrammetric Computer Vision, Machine Learning, Optimization, Signal Processing, Statistics.

Bachelor of Technology in Mechanical Engineering

Sastra Deemed to be University

Jul 2017 – May 2021

Thanjavur, India

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, Java, SQL
Computer Vision	3D Geometry, Camera Models, 3D-to-2D Projection, Point-Cloud Processing, Image Segmentation, Object Detection, Coordinate Transformations
Sensor Fusion	LiDAR-Camera Fusion, Multi-Sensor Data Association, Kalman Filtering, Motion Models, Extended Kalman Filter, State Estimation, Multi-Object Tracking
Robotics	ROS 2, ROS Bags, RViz, Real-Time Perception
ML & Computing	OpenCV, PyTorch, TensorFlow, NumPy, SciPy, CUDA
Tools	Linux, Git, GitLab, Docker

LANGUAGES

German (B1), English (C2), Telugu (C2)